

Integrative Identification and Independent Validation of miRNA Biomarkers Associated with Triple-Negative Breast Cancer Using TCGA and GEO

Independent Research Project

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Background:

MicroRNAs (miRNAs) are small non-coding RNAs that regulate gene expression and influence cancer-related processes such as proliferation, apoptosis, migration, and invasion. Altered miRNA expression has therefore been investigated as a potential source of diagnostic and disease-associated biomarkers in breast cancer.

However, many reported miRNA candidates are derived from individual datasets, and their reproducibility across independent cohorts remains an important challenge. Publicly available datasets from The Cancer Genome Atlas (TCGA) and the Gene Expression Omnibus (GEO) provide an opportunity to systematically identify and independently validate TNBC-associated miRNAs.

In this study, TCGA-BRCA miRNA expression data were analyzed to identify miRNAs differentially expressed in clinically defined TNBC compared with normal breast tissue and other breast cancer receptor groups. Candidate miRNAs were subsequently evaluated using independent GEO datasets, followed by ROC analysis, survival analysis, and miRNA–target functional analysis. The overall aim was to identify reproducible candidate miRNA biomarkers associated with TNBC through an integrative bioinformatics approach.

Research Question

Can reproducibly dysregulated miRNAs be identified as candidate biomarkers associated with Triple-Negative Breast Cancer through integrative analysis of TCGA and independent GEO datasets?

Keywords

TNBC · miRNA · Biomarkers · TCGA · GEO · Bioinformatics · DESeq2 · limma

Aim

To identify and independently validate miRNAs associated with Triple-Negative Breast Cancer using publicly available TCGA and GEO datasets.

Objectives

- 1) To identify differentially expressed miRNAs associated with TNBC using TCGA-BRCA miRNA expression data and comparisons with normal breast tissue and other breast cancer subtypes.
- 2) To independently validate prioritized miRNA candidates using GEO datasets and evaluate their potential for TNBC classification through ROC analysis.
- 3) To investigate the biological and clinical relevance of prioritized miRNAs using survival analysis, miRNA–target interactions, and functional enrichment analysis.

Methodology:

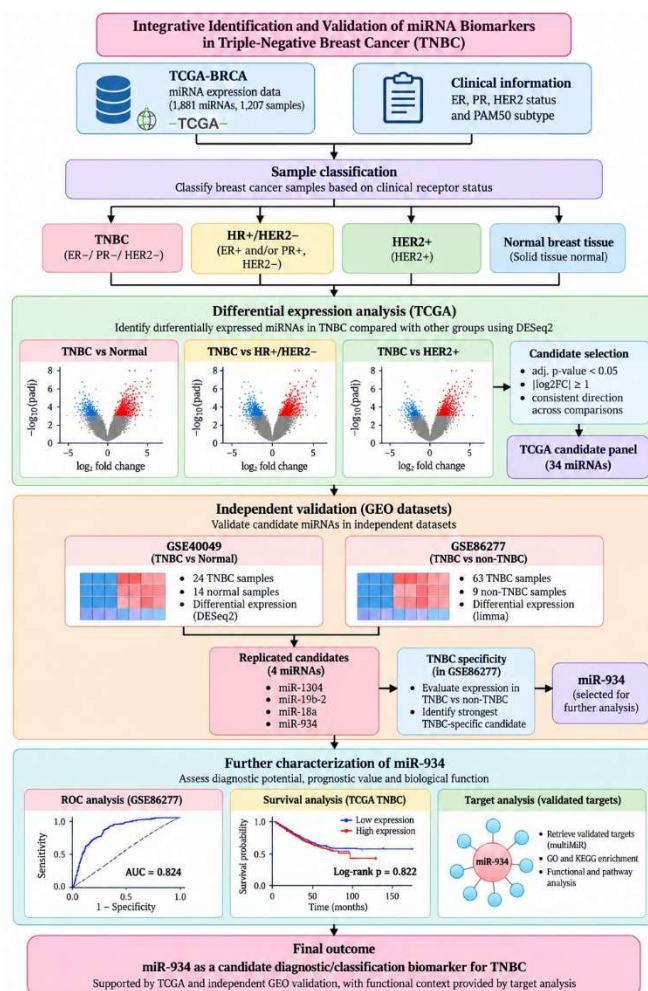


Figure 1 Methodology and workflow

An integrative bioinformatics workflow was used to identify and validate candidate miRNA biomarkers associated with triple-negative breast cancer (TNBC). TCGA-BRCA miRNA expression data were obtained from the GDC, together with clinical information including estrogen receptor (ER), progesterone receptor (PR), and HER2 status. Samples were clinically classified into TNBC (ER-/PR-/HER2-), HR+/HER2-, HER2+ and normal groups. Differential miRNA expression was evaluated using DESeq2, with candidates selected using adjusted $p < 0.05$, $|\log_2FC| \geq 1$ and consistent direction of change across comparisons.

The resulting TCGA candidates were independently evaluated using two GEO datasets. GSE40049 was analysed to assess reproducibility between TNBC and normal breast tissue, while GSE86277 was used to assess TNBC specificity against non-TNBC samples. RNA-sequencing count data were analysed using DESeq2, whereas microarray data from GSE86277 were analysed using limma. Candidates showing consistent differential expression in independent datasets were prioritized, leading to the selection of miR-934 for further characterization.

The discriminatory potential of miR-934 was evaluated using ROC curve and AUC analysis. Its potential prognostic relevance was assessed in TCGA TNBC patients using Kaplan–Meier survival analysis and Cox proportional-hazards regression. Finally, experimentally supported miR-934 target genes were retrieved using multiMiR and explored through target-gene enrichment and network analysis to provide functional biological context.

Results

4.1 TCGA-BRCA miRNA profiling and clinical subgrouping

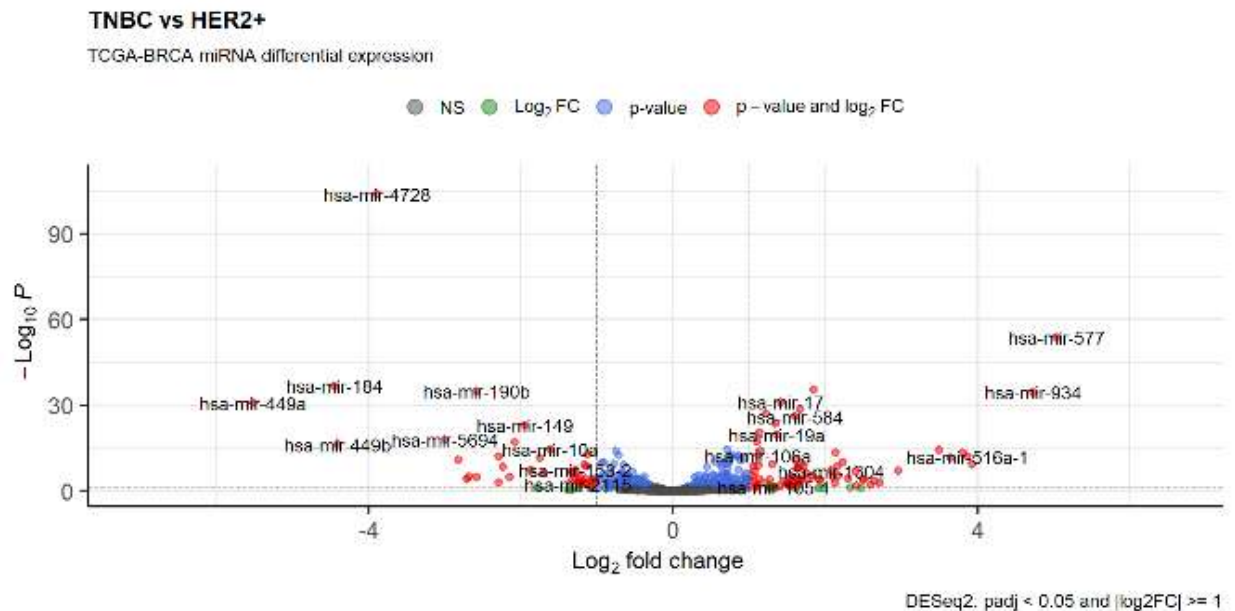
TCGA-BRCA miRNA expression data consisted of **1,881 miRNA features across 1,207 samples**. Clinical annotation based on ER, PR and HER2 status identified **133 TNBC, 542 HR+/HER2– and 125 HER2+ samples** before restricting to one tumor sample per patient for differential-expression analysis.

Importantly, clinical TNBC classification was used rather than relying solely on PAM50 Basal classification, because the two classifications were not completely concordant.

4.2 Differential miRNA expression in TCGA

DESeq2 analysis identified **218 significant miRNAs** in TNBC vs Normal, **144** in TNBC vs HR+/HER2–, and **114** in TNBC vs HER2+ using adjusted $p < 0.05$ and $|\log_2FC| \geq 1$.

Across the subtype comparisons, **89 miRNAs** were common, with **88 showing consistent expression direction**. Incorporating the TNBC vs Normal comparison resulted in **38 common miRNAs**, of which **34 showed consistent direction of expression**. These 34 miRNAs were selected for subsequent independent validation.



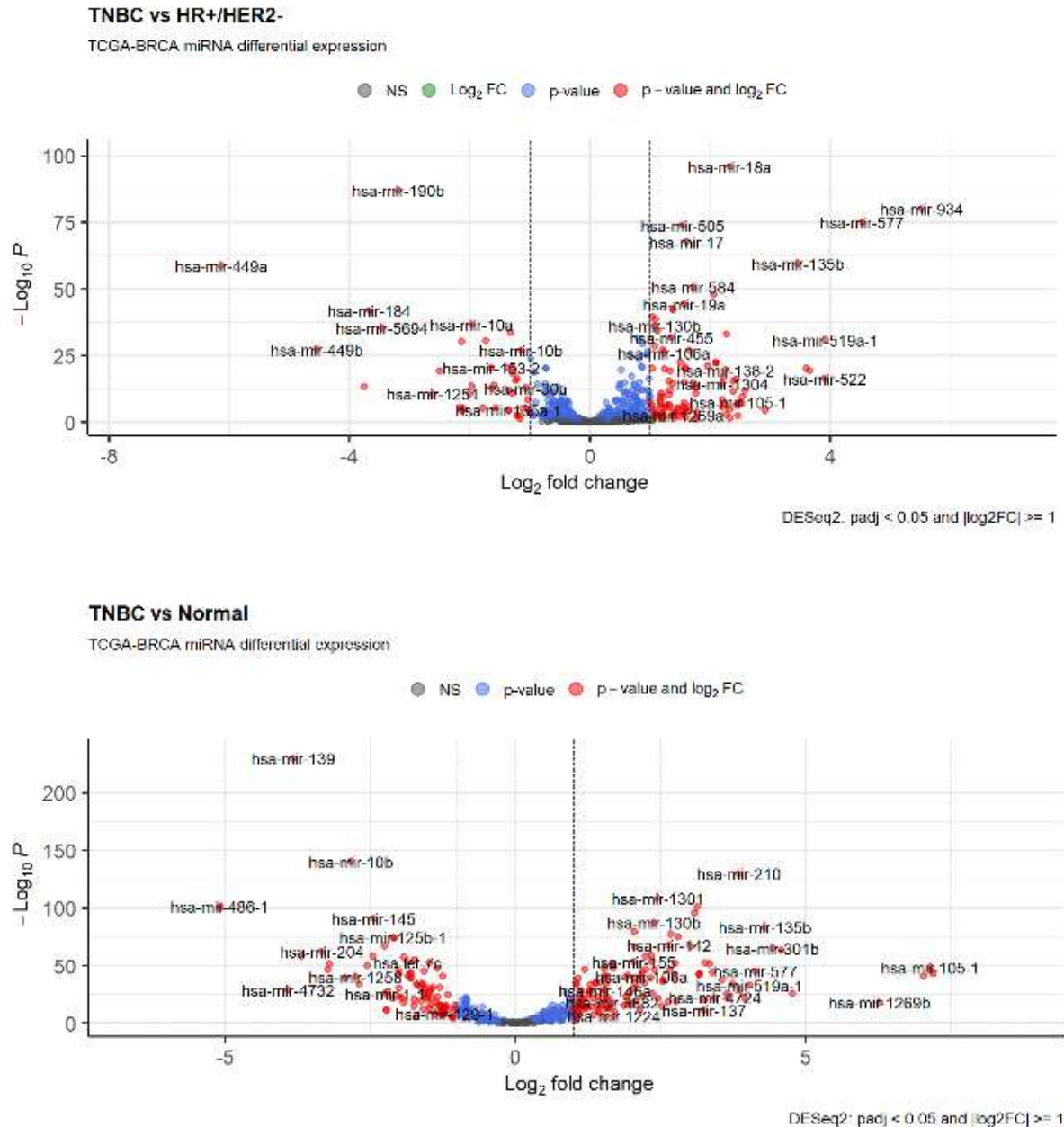


Figure 2 Differential expression of miRNAs in TCGA-BRCA.

4.3 Candidate miRNA selection

The 34 directionally consistent candidates were ranked according to their minimum absolute log₂ fold-change across the three TCGA comparisons.

This produced a prioritized candidate set including miR-577, miR-522, miR-516a, miR-934, miR-519a, miR-138, miR-137 and others.

The precursor miRNA identifiers were subsequently mapped to mature miRNA forms using miRNA annotation resources, generating **55 unique mature miRNA identifiers**.

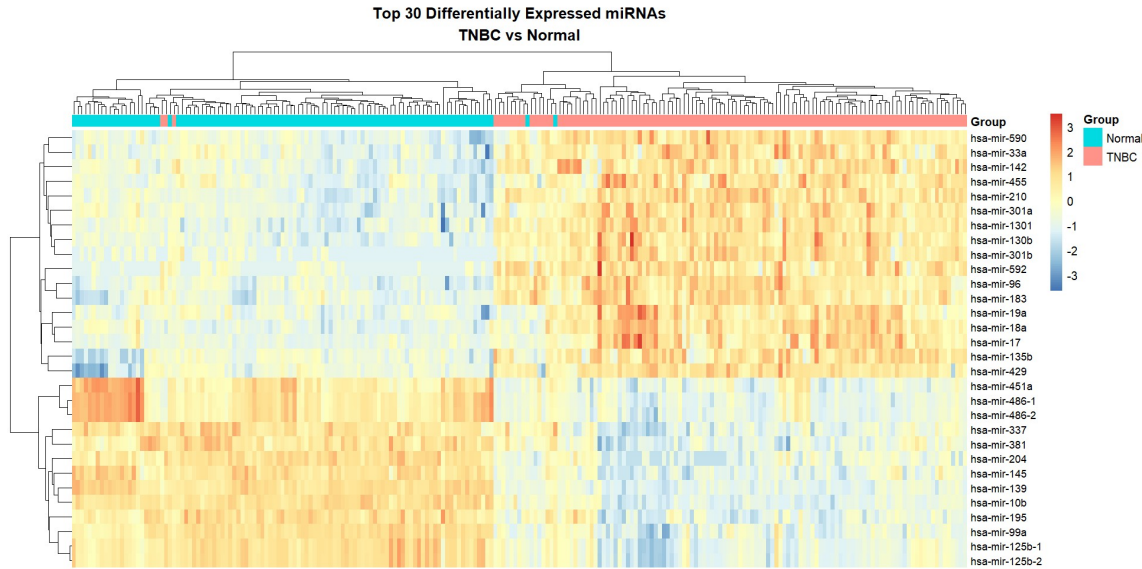


Figure 3 Heatmap of differentially expressed miRNAs in TNBC.

4.4 Independent validation using GSE40049

Independent validation was performed using **GSE40049**, containing **24 TNBC** and **14 adjacent-normal samples**.

After processing the small-RNA sequencing count files and filtering low-abundance features, **870 miRNAs** were retained for differential-expression analysis.

Of the TCGA candidate set, **30 candidates were directly represented in the GSE40049 dataset**. Four candidates met the predefined replication criteria:

- adjusted p-value < 0.05
- $|\log_2FC| \geq 1$
- consistent direction with TCGA

The four replicated candidates were:

Candidate	TCGA log2FC	GEO log2FC	GEO adjusted p
miR-1304	+1.54	~+2.88	4.16×10^{-6}
miR-19b-2	+1.21	~+1.59	0.00473
miR-18a	+2.80	~+1.35	0.0395
miR-934	+3.56	~+1.79	0.0338

Thus, **miR-934** was among the candidates independently replicated in a separate TNBC-versus-normal dataset.

4.5 TNBC specificity using GSE86277

The second independent dataset, **GSE86277**, was used to evaluate whether the replicated candidates were associated specifically with TNBC rather than breast cancer in general.

The downloaded series matrix contained **63 TNBC and 9 non-TNBC samples**. Because this was microarray expression data, differential expression was evaluated using limma rather than DESeq2.

Among the four replicated candidates, **miR-934 showed the strongest evidence of TNBC enrichment**, with:

logFC = +2.873, adjusted p = 0.0104

miR-18a showed the same direction but did not reach FDR significance, while miR-1304 and miR-19b-2 did not demonstrate significant TNBC-specific differences in this dataset.

These results prioritized **miR-934** for further characterization.

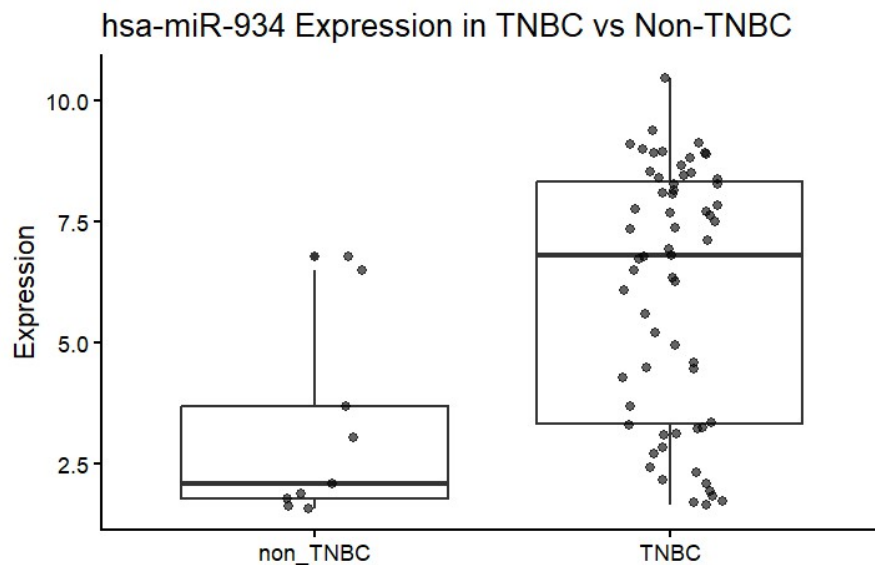


Figure 4 miR-934 expression in TNBC and non-TNBC samples in GSE86277.

4.6 Diagnostic/classification performance of miR-934

ROC analysis in GSE86277 produced:

AUC = 0.824

95% CI = 0.689–0.959

This indicates that miR-934 demonstrated moderate-to-good discriminatory ability between TNBC and non-TNBC samples in this independent dataset.

However, because only **9 non-TNBC samples** were available in the downloaded dataset, this result should be considered **supportive rather than definitive clinical diagnostic validation**.

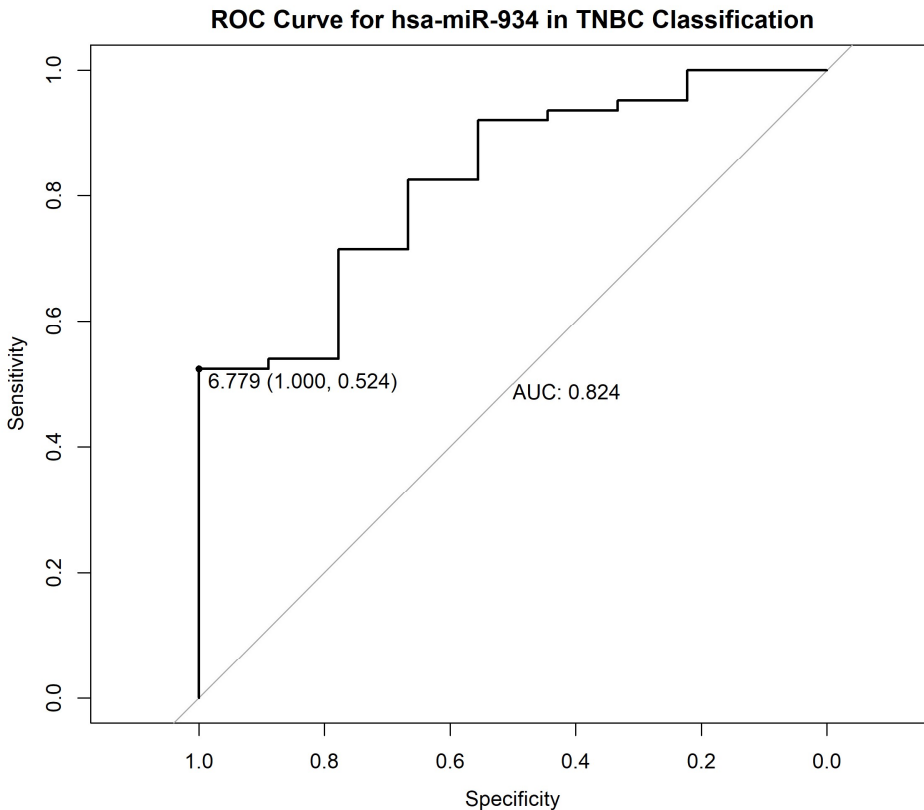


Figure 5 ROC curve demonstrating the discriminatory performance of miR-934 for TNBC classification.

4.7 Survival analysis

Survival analysis was performed in **135 TCGA TNBC patients**, including **24 observed deaths**.

Patients were divided into high- and low-miR-934 groups using the median expression.

Kaplan–Meier analysis showed **no significant difference in overall survival**:

Log-rank $p = 0.822$

Cox regression similarly showed:

HR = 0.912

95% CI = 0.401–2.075

$p = 0.827$

Therefore, miR-934 was **not supported as a prognostic biomarker for overall survival in this analysis**.

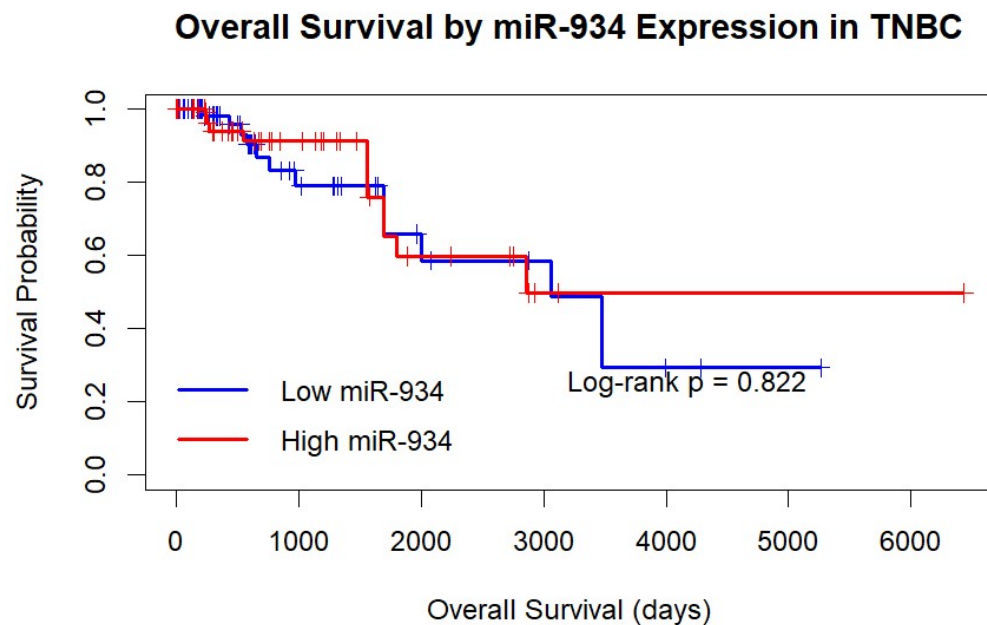


Figure 6 Kaplan–Meier overall-survival analysis according to miR-934 expression.

4.8 miR-934 target analysis

Validated miR-934 target interactions were subsequently investigated to provide biological context for the candidate biomarker.

The target-analysis stage identified experimentally supported target genes and was followed by functional enrichment and network analysis.

The target network was treated as **supporting biological interpretation**, rather than evidence that these targets establish the diagnostic function of miR-934.

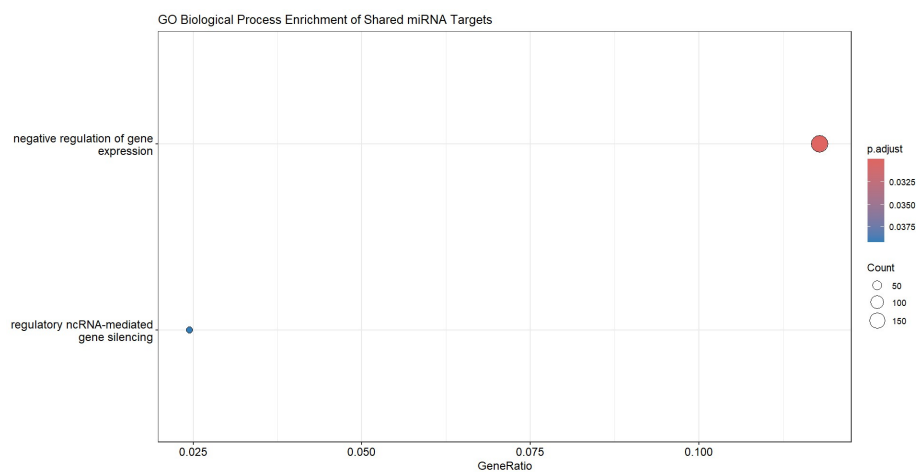


Figure 7 miR-934 GO enrichment dot plot

Validated Target Network of hsa-miR-934

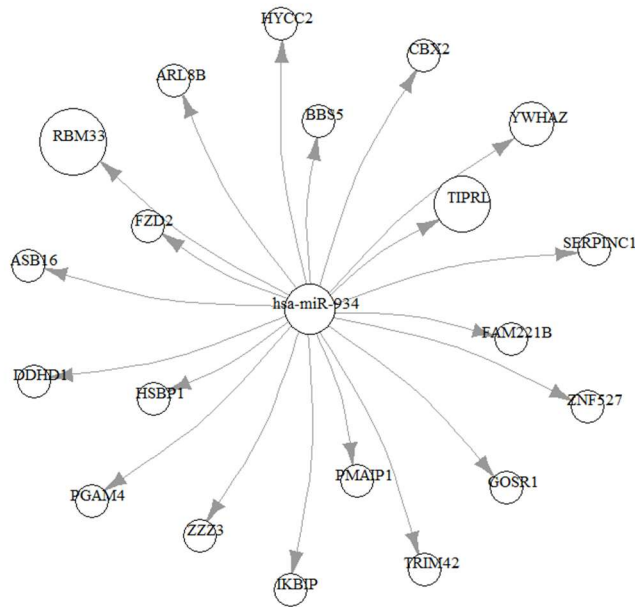


Figure 8 Experimentally supported miR-934 target network.

Conclusion:

An integrative analysis of TCGA-BRCA and independent GEO datasets identified miR-934 as a reproducibly upregulated and TNBC-enriched candidate miRNA. miR-934 demonstrated independent TNBC discrimination in GSE86277 (AUC = 0.824), while survival analysis did not support a significant association with overall survival. These findings support miR-934 as a candidate TNBC classification/diagnostic biomarker warranting further validation.

Discussion

The present study used an integrative computational approach to identify and evaluate miRNA candidates associated with triple-negative breast cancer (TNBC). Instead of relying on a single comparison, clinically defined TNBC samples were compared with normal breast tissue, HR+/HER2- breast cancer and HER2+ breast cancer. This allowed candidates associated with TNBC to be prioritized based on both differential expression and consistency across breast cancer subtypes.

The TCGA analysis initially identified a broad set of differentially expressed miRNAs. After applying effect-size and direction-consistency criteria across the three comparisons, 34 candidates remained. Independent analysis of GSE40049 further reduced this list to four candidates: miR-1304, miR-19b-2, miR-18a and miR-934.

Among these candidates, miR-934 showed the strongest evidence for TNBC association in the second independent GEO dataset, GSE86277. Its significantly higher expression in TNBC compared with non-TNBC samples, together with an AUC of approximately 0.824, suggests that miR-934 may have discriminatory potential for TNBC classification.

However, survival analysis did not demonstrate a significant association between miR-934 expression and overall survival. Therefore, the present findings do not support miR-934 as a prognostic biomarker. Instead, its potential utility in this study is more consistent with **TNBC classification/diagnostic discrimination**.

Target analysis provided additional biological context by examining experimentally supported molecular targets of miR-934. These results can help generate hypotheses regarding pathways through which miR-934 may be involved in TNBC biology, but target enrichment alone does not establish a mechanistic role.

Limitations

You should explicitly include these. They make the project look more scientifically mature.

1. Retrospective public datasets

The study is entirely computational and based on publicly available datasets. Experimental validation was not performed.

2. Limited independent validation sample size

The GSE86277 analysis contained only **9 non-TNBC samples**, which limits the reliability and generalizability of the ROC estimate.

3. Platform differences

TCGA and the GEO datasets were generated using different technologies and processing pipelines. TCGA used sequencing-based expression quantification, whereas GSE86277 used a microarray platform.

4. Survival analysis had relatively few events

Only **24 deaths** were available among the 135 TNBC patients used for survival analysis. Consequently, the absence of a significant survival association should not be interpreted as definitive evidence that miR-934 has no prognostic relevance.

5. No experimental validation

The target-gene analysis was based on existing validated interactions. The biological effects of miR-934 and its targets in TNBC were not experimentally confirmed.

6. miR-934 is not a newly discovered biomarker

Previous research has reported an association between miR-934 and TNBC. Therefore, the contribution of this project should be presented as **integrative computational identification and independent validation**, rather than claiming discovery of a novel biomarker.

Overall, the integrative analysis identified miR-934 as a reproducibly upregulated TNBC-associated miRNA with discriminatory potential across independent datasets. Although its prognostic significance was not supported by overall-survival analysis, the findings provide computational evidence supporting further investigation of miR-934 as a candidate biomarker for TNBC classification.

References

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